

KARGOPOL', Russian FederationWMO: **228450**Lat: **61.5067N**Lon: **38.9347E**Elev: **413**StdP: **14.48**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-16.1	-26.7	1.3	-22.2	-20.5	1.8	-16.2	16.3	23.0	14.4	23.7	2.0	20	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	81.5	67.0	78.1	65.6	74.7	63.5	69.7	78.3	67.4	75.3	65.4	72.5	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	99.4	74.2	64.4	92.1	71.2	62.3	85.4	69.8	33.9	78.9	32.1	75.4	30.5	72.5	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.8	11.6	DB	-25.8	85.7	7.4	3.7	-31.1	88.4	-35.4	90.6	-39.5	92.7	-44.9	95.4	(4)
(5)			WB	-26.0	72.3	7.3	3.2	-31.3	74.6	-35.5	76.5	-39.6	78.2	-45.0	80.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	12.0	13.8	23.2	36.3	48.5	57.5	63.4	58.4	49.2	37.5	25.5	18.3	(6)	
(7)		DBStd	20.18	14.95	14.07	9.43	8.42	9.28	7.77	6.83	6.53	6.81	7.34	10.85	13.19	(7)	
(8)		HDD50	5822	1179	1014	832	419	143	18	2	8	95	393	736	983	(8)	
(9)		HDD65	10299	1644	1434	1297	861	515	250	113	223	474	854	1186	1448	(9)	
(10)		CDD50	1104	0	0	0	8	96	242	416	267	71	4	0	0	(10)	
(11)		CDD65	107	0	0	0	0	4	24	62	17	0	0	0	0	(11)	
(12)		CDH74	869	0	0	0	2	44	229	464	130	0	0	0	0	(12)	
(13)		CDH80	165	0	0	0	0	4	49	91	21	0	0	0	0	(13)	
(14)	Wind	WSAvg	5.4	5.6	5.8	5.7	5.2	5.5	5.0	4.5	4.5	5.0	5.8	6.0	6.2	(14)	
(15)	Precipitation	PrecAvg	24.3	1.3	1.0	1.2	1.5	2.0	2.8	2.8	3.3	2.4	2.4	1.9	1.7	(15)	
(16)		PrecMax	31.4	2.4	2.1	2.6	3.5	4.9	5.3	5.5	10.0	4.4	4.6	3.5	3.1	(16)	
(17)		PrecMin	16.4	0.3	0.1	0.1	0.2	0.6	0.2	0.9	0.8	0.5	0.1	0.4	0.4	(17)	
(18)		PrecStd	3.2	0.4	0.5	0.7	0.8	1.0	1.3	1.1	1.9	1.0	1.0	0.7	0.7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	36.6	46.3	68.1	79.7	85.6	86.9	82.8	71.2	58.7	45.1	40.6	(19)	
(20)			MCWB	33.8	34.9	37.4	53.1	61.0	68.5	70.9	69.5	62.0	52.8	43.5	39.6	(20)	
(21)		2%	DB	33.9	34.6	40.8	60.1	73.6	80.0	82.3	77.4	66.2	53.2	42.3	36.7	(21)	
(22)			MCWB	32.7	33.2	34.7	48.5	57.7	65.2	68.3	65.4	59.5	50.3	40.8	35.8	(22)	
(23)		5%	DB	32.7	33.1	37.5	53.6	68.7	75.9	79.1	73.2	62.3	50.0	39.8	34.6	(23)	
(24)			MCWB	31.9	32.0	33.6	44.3	55.5	63.3	67.3	63.7	56.7	48.1	38.9	33.6	(24)	
(25)		10%	DB	31.2	31.5	35.2	49.1	64.6	71.8	75.9	69.8	59.3	47.3	37.5	33.1	(25)	
(26)			MCWB	30.4	30.5	32.6	41.6	53.2	60.9	65.4	61.6	55.2	45.7	36.7	32.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	35.3	38.7	54.4	64.4	71.0	73.2	71.3	63.3	53.8	44.2	39.8	(27)	
(28)			MCDWB	34.8	36.3	43.8	67.9	74.4	81.8	82.6	79.6	69.0	57.1	44.6	40.3	(28)	
(29)		2%	WB	33.0	33.5	36.1	49.1	60.6	67.7	70.6	67.2	60.3	50.5	41.0	36.0	(29)	
(30)			MCDWB	33.6	34.2	39.4	57.9	70.1	77.6	79.5	74.3	64.8	52.4	41.9	36.7	(30)	
(31)		5%	WB	32.0	32.2	34.3	45.7	57.4	65.0	68.5	64.8	57.8	48.4	38.9	33.7	(31)	
(32)			MCDWB	32.7	33.0	36.6	52.4	66.0	73.5	76.4	70.8	61.3	50.1	39.7	34.3	(32)	
(33)		10%	WB	30.2	30.5	33.0	42.3	54.5	62.1	66.7	62.7	55.7	45.8	36.7	32.3	(33)	
(34)			MCDWB	31.0	31.6	35.0	48.5	63.3	69.5	74.0	68.6	58.7	47.1	37.3	33.1	(34)	
(35)	Mean Daily Temperature Range			MDBR	10.6	12.2	14.7	16.1	17.6	16.9	16.4	15.0	12.2	7.6	7.2	8.6	(35)
(36)		5% DB	MCDBR	8.2	8.0	15.5	23.7	24.2	22.7	21.0	20.2	17.3	10.9	7.2	5.7	(36)	
(37)			MCWBR	8.3	7.7	11.2	14.1	11.6	10.7	9.6	9.7	10.3	8.3	7.0	5.7	(37)	
(38)		5% WB	MCDWB	8.7	8.1	11.6	21.4	20.7	19.9	18.6	17.2	14.2	10.5	6.8	5.7	(38)	
(39)			MCWBR	8.7	7.9	8.7	13.5	11.2	10.4	9.7	9.3	9.3	8.4	6.7	5.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.247	0.295	0.321	0.361	0.352	0.351	0.386	0.369	0.337	0.318	0.269	0.216	(40)	
(41)		taud		2.023	2.077	2.120	2.241	2.443	2.472	2.386	2.412	2.488	2.389	2.127	1.956	(41)	
(42)		Ebn at Noon		140	206	247	261	276	278	263	258	247	206	131	94	(42)	
(43)		Edh at Noon		13	24	32	35	31	31	33	30	23	18	11	8	(43)	
(44)	All-Sky Solar Radiation	RadAvg		80	262	659	1135	1563	1673	1634	1192	680	269	76	37	(44)	
(45)		RadStd		10	64	96	120	117	136	172	108	71	33	13	5	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.22	N/A	N/A	N/A	N/A	N/A	-389	-457	N/A	N/A
(47)	Regional (6 neighbors)	+1.15	N/A	N/A	N/A	N/A	N/A	-356	-391	N/A	N/A

Nomenclature: See separate page